12.3 Surface Areas of Pyramids and Cones

Name _____ Period ____ Date _____

Geometry 12.3 (30-minute plan) · Student sheet

I can…

- find the surface area of a regular pyramid.
- find the surface area of a right cone.
- find a slant height with the Pythagorean Theorem.

Vocabulary · fill in the missing word

Regular pyramid — a pyramid whose base is a regular polygon and whose lateral faces are _____ isosceles triangles

Slant height — the height of a lateral face of a regular pyramid, or the _____ from the vertex of a cone to the edge of its base

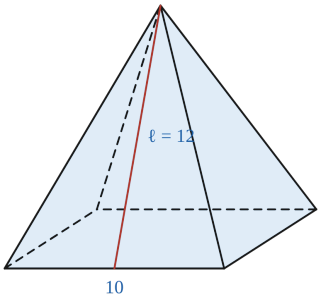
Lateral surface of a cone — the _____ surface of a cone, between the base and the vertex

Part 1 · Notes and examples (fill in each note, then work the example)

Surface area of a regular pyramid: $S = B +$ _____, where P is the perimeter of the base and ℓ is the slant height.

Example 1

Find the surface area of the regular square pyramid.

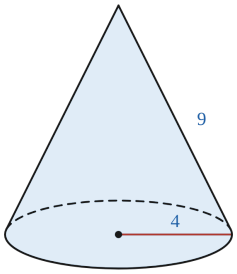


Finding a slant height: the height, half a side of the base (or the radius), and the slant height form a _____.

Surface area of a right cone: $S = \pi r^2 +$ _____

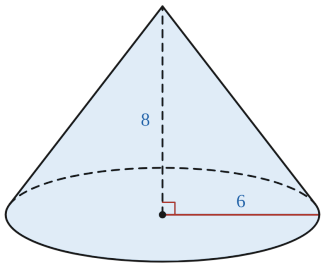
Example 2

Find the surface area of the cone. Round to the nearest hundredth.



Example 3

A cone has a radius of 6 and a height of 8. Find its surface area. Round to the nearest hundredth.



Part 2 · Partner practice

Partner practice

A cone has a radius of 5 and a height of 12. Find its surface area. Round to the nearest hundredth.